

# Shaoyuan Xie

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## EDUCATION

### University of California, Irvine

*Ph.D. in Computer Science, Advisor: Professor Qi Alfred Chen*

Irvine, CA

*Expected Sep 2027*

### Huazhong University of Science and Technology (HUST)

*B.Eng. in Automation, GPA: 3.97 (Top 1st / 212 students)*

Wuhan, CN

*Jun 2023*

## INDUSTRY EXPERIENCE

### Waymo

*Research Intern, AI Foundation | Mentor: Chen Song | Manager: Mingxing Tan*

Mountain View, CA

*Jun 2026 – Sep 2026*

- Conduct research on a Gemma-based multimodal perception foundation model with camera and LiDAR inputs, improving spatial perception and reasoning.
- Perform large-scale JAX training on 256 TPUs using FSDP, activation rematerialization, and tensor parallelism.

### Mercedes-Benz Research & Development North America

*Research Intern, AD Perception | Mentor: Thomas Monninger | Manager: Sihao Ding*

San Jose, CA

*Jun 2025 – Sep 2025*

- Semantic distillation from LLM to BEV encoder for safer end-to-end driving.

### Shanghai Artificial Intelligence Laboratory

*Research Intern, OpenMMLab | Mentor: Wenwei Zhang | Manager: Kai Chen*

Shanghai, CN

*Jun 2023 – Aug 2023*

- Post-training of [InternLM](#) language models.

## SELECTED PUBLICATIONS

**10 top-tier publications**, including **6 first/co-first-author papers** at TPAMI, ICLR, NeurIPS, ICCV, ECCV, CoRL, IROS and NDSS

### • SafeDriveVLA: Navigation-Conditioned World Model Dreaming for Conflict-Aware End-to-End Autonomous Driving

Shaoyuan Xie\*, Zihan Zhang\*, Jingxuan Wang, Jiashu Qu, Xiaoqing Liang, Lingdong Kong, Junchi Lu, Henrik I Christensen, Qi Alfred Chen  
*CoRL, 2026*

### • BEVLM: Distilling Semantic Knowledge from LLMs into Bird’s-Eye View Representations

Thomas Monninger\*, Shaoyuan Xie\* (\*order determined alphabetically), Qi Alfred Chen, Sihao Ding  
*ECCV, 2026*

### • FlyTrap: Physical Distance-Pulling Attack Towards Camera-based Autonomous Target Tracking Systems

Shaoyuan Xie, Mohamad Fakhri, Junchi Lu, Fayzah Alshammari, Ningfei Wang, Takami Sato, Halima Bouzidi, Mohammad Al Faruque, Qi Alfred Chen  
*NDSS, 2026*

### • Are VLMs Ready for Autonomous Driving? An Empirical Study from Reliability, Data, and Metric Perspectives

Shaoyuan Xie, Lingdong Kong, Yuhao Dong, Chonghao Sima, Wenwei Zhang, Qi Alfred Chen, Ziwei Liu, Liang Pan  
*ICCV, 2025*

### • Can We Trust Embodied Agents? Exploring Backdoor Attacks against Embodied LLM-Based Decision-Making Systems

Ruochen Jiao\*, Shaoyuan Xie\* (\*co-first author), Justin Yue, Takami Sato, Lixu Wang, Yixuan Wang, Qi Alfred Chen, Qi Zhu  
*ICLR, 2025*

- **Benchmarking and Improving Bird's Eye View Perception Robustness in Autonomous Driving**  
Shaoyuan Xie, Lingdong Kong, Wenwei Zhang, Jiawei Ren, Liang Pan, Kai Chen, Ziwei Liu  
*TPAMI, 2025*
- **On the Adversarial Robustness of Camera-based 3D Object Detection**  
Shaoyuan Xie, Zichao Li, Zeyu Wang, Cihang Xie  
*TMLR, 2024*
- **The Heat is On: Understanding and Mitigating Vulnerabilities of Thermal Image Perception in Autonomous Systems**  
Hrushikesh Varma, Shaoyuan Xie, Michael Clifford, Qi Alfred Chen, Takeshi Sugawara, Sara Rampazzi  
*NDSS, 2026*
- **Revisiting Physical-World Adversarial Attack on Traffic Sign Recognition: A Commercial Systems Perspective**  
Ningfei Wang, Shaoyuan Xie, Takami Sato, Yunpeng Luo, Kaidi Xu, Qi Alfred Chen  
*NDSS, 2025*
- **RoboDepth: Robust Out-of-Distribution Depth Estimation under Corruptions**  
Lingdong Kong, Shaoyuan Xie, Hanjiang Hu, Lai Xing Ng, Benoit R. Cottureau, Wei Tsang Ooi  
*NeurIPS, 2023*
- **Adversarial Spatial Transform Marker: A Black-Box Pose Estimation Attack for Autonomous UAV Landing**  
Junchi Lu, Fayzah Alshammari, Shaoyuan Xie, Xiaoqing Liang, Qi Alfred Chen  
*IROS, 2026*
- **"Prompter Says": A Linguistic Approach to Understanding and Detecting Jailbreak Attacks Against Large-Language Models**  
Dylan Lee, Shaoyuan Xie, Shagoto Rahman, Kenneth Pat, David Lee, Qi Alfred Chen  
*ACM CCS Workshop - LAMPS, 2024*

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## RESEARCH IMPACT

- Email acknowledgements from DJI on autonomous tracking vulnerabilities
- Responsible vulnerability disclosure acknowledged by a major automotive manufacturer for traffic-sign recognition vulnerabilities

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## AWARDS AND HONORS

- Best Demo, 3rd USENIX Symposium on Vehicle Security and Privacy (2025, *selected for best demo rate 18.2% = 2/11*)
- Honours Bachelor's Degree (2023, *top 1% in univ.*)
- Outstanding Undergraduate Student (2021, *top 1% in univ.*)
- National Scholarship, Ministry of Education of China (2021, *top 0.2% nation-wide*)
- National Scholarship, Ministry of Education of China (2020, *top 0.2% nation-wide*)

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## ACADEMIC SERVICES

- *Competition Organizer*: IROS RoboSense (2025), ICRA RoboDrive (2024), ICRA RoboDepth (2023)
- *PC member*: ACM CCS Artifact Evaluation (2024, 2025, 2026), NDSS Artifact Evaluation (2025)
- *Reviewer*: CVPR (2023, 2024, 2025, 2026), ICCV (2025), ECCV (2026), BMVC (2026), ICLR (2026), CoRL (2026), IROS (2026), IJCV, RA-L, TDSC; TMM, TMLR
- *External Reviewer*: USENIX Security (2024, 2025), IEEE S&P (2024), ACM CCS (2024)